Stormbird Instruction Manual

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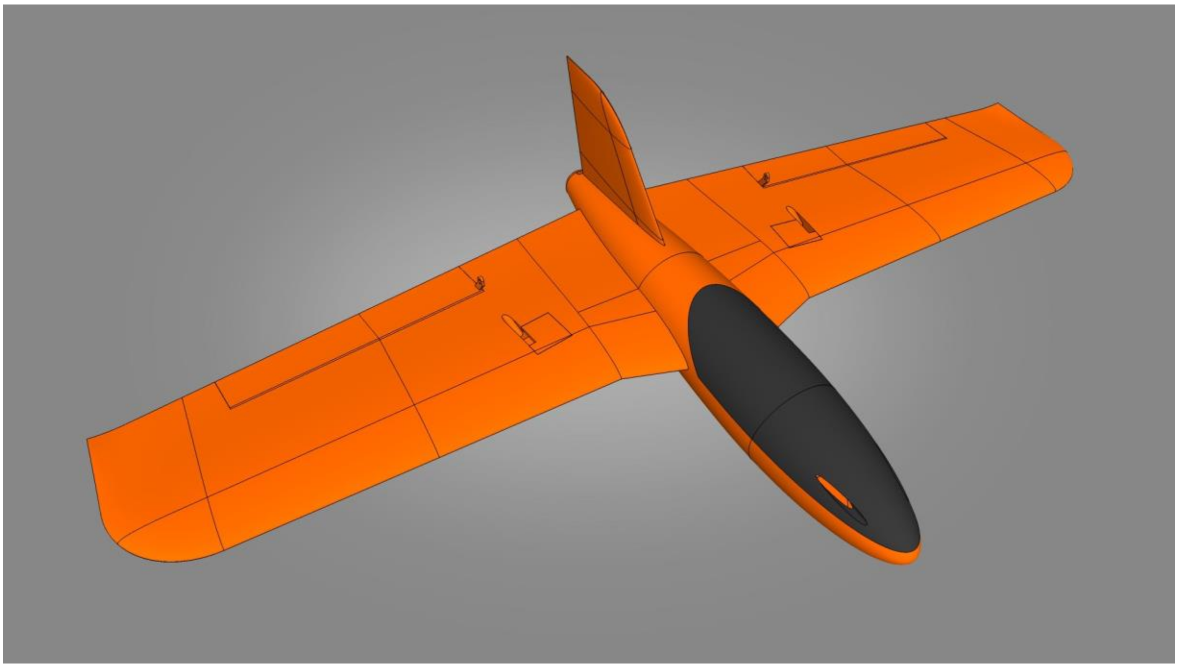


The Stormbird is a 1.1m plank style flying wing created by Stunt Double

This is a fast and exciting RC plane that has been developed for FPV flying. It's free to download from Thingiverse. The Stormbird has a slippery profile and is built for speed & efficiency. It offers excellent flight characteristics with its slim semi-symmetrical reflexed airfoil. It features a long spacious fuselage with a secure battery attachment plate and has a magnetic slide on canopy for quick easy access.

www.thingiverse.com/thing:6174038

SPECIFICATIONS	
Wingspan	110cm
Length	58cm
Printed parts weight	600-650g
Flying weight	900-1200g
Airfoil	PW75
Wing area	$20dm^2$
Wing loading	$45\text{-}60g/dm^2$

Equipment

Hardware & supplies

- Filament PLA+ 600-650g
- Carbon fiber tube 6mm x 1m (1)
- Carbon fiber tube 6mm x 300mm (1)
- Carbon fiber rod 3mm x 330mm (2)
- Carbon fiber rod 3mm x 30mm (3)
- Magnet round 6mm x 3mm (4)
- Hex head screw M3 x 10mm (1)
- Locknut M3 (1)
- Push rod, clevis, ball link 2mm (2)
- Battery strap 20mm x 200mm (1)
- Velcro adhesive strip 25mm x 150mm (1)
- Medium CA
- UHU Por / E6000
- Sandpaper

Servos

Two 12-13g mini sized metal geared servos are required. Measuring approx. 24x11.6x24mm. The Emax ES08MDII digital servo is recommended.

Motor

A 28mm O.D. outrunner with a 19/16mm bolt pattern is required. Weight between 35-70g. The T-motor F80 PRO 2500kv (on 4S with a 6x4 prop) is recommended for a lightweight setup or the SunnySky X Series V3 X2216 1400kv (on 4S with an 8x6 prop) for a more efficient FPV setup.

Propellor

Use a 6-9 inch prop.

ESC

It is recommended to use a small lightweight BLHeli 40-50A ESC such as the Flycolor Raptor 5 40A.

Battery

A 4S 1500-3000 mAh Li-po or Li-ion is required. Weight range from 150-300g. The CNHL Black Series 4S 1500 mAh 100C is recommended for a short duration LOS flying setup or the CNHL Black Series 4S 2200 mAh 40C for a longer flying FPV setup.

Flight controller

Use a small compact flight controller. The flight controller tray has 30.5 x 30.5mm or 20 x 20mm mounting holes. The F405-TE or F405-WMN boards from Matek are suitable.

Printing

Filament

It is recommended to use PLA+ (such as eSUN brand). Regular PLA is not suitable, PLA+ is much stronger. PETG, ABS or LW-PLA filaments could also be used.

Print bed

The 3D printed parts will fit onto a print bed no smaller than 200x200x200mm. This will be suitable for common printers such as the Creality Ender 3/5, Prusa MK3/4 & Bambu Lab range.

Layers, lines, walls & infill

A layer height of 0.2 to 0.28 can be used. Thin layers will give a better appearance, however using thicker layers will significantly speed up printing times on older slower printers.

A line width of 0.4 to 0.42 should be used.

Most of the parts should be printed using a single wall with a small amount of infill density, typically around 4% with the cubic pattern. The battery plate, flight controller tray, motor mount & servo trays should be printed solid with 3 walls and 50% infill density.

The elevon inner parts should be printed with 3mm of bottom layers to create a solid control horn (eg. 0.2 x 15 layers).

Support & brim

The canopy front, nose & wing inners should be printed with support. All other parts do not require support.

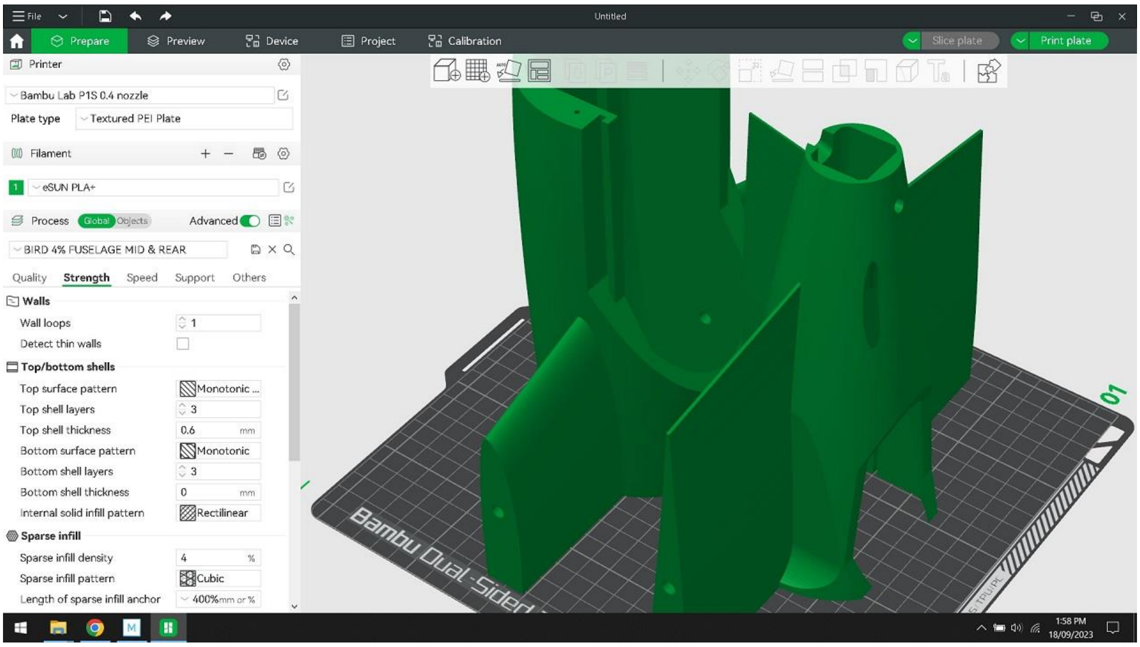
A brim should be used on taller/wider parts to prevent warping and bed detachment.

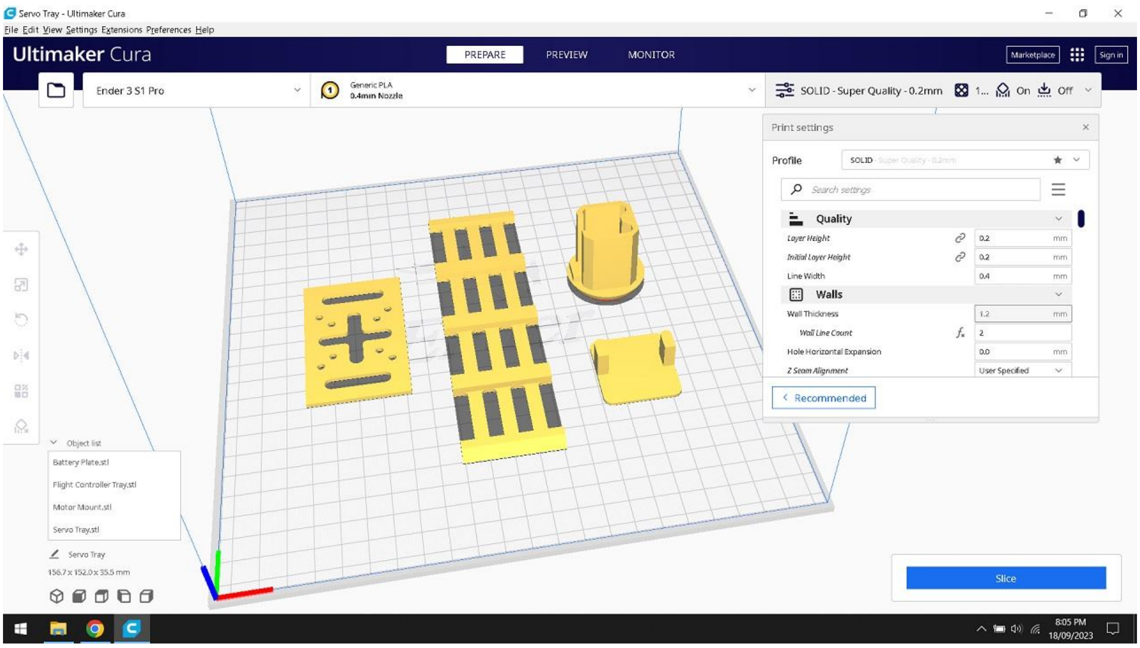
Other slicer settings

Pay attention to the Z seam position for each part. The Z seam is quite noticeable on single wall prints so it's best to hide it as much as possible.

Increasing the printing temperature & reducing the fan speed can help improve the strength of single wall PLA+ parts.

Settings for the nozzle & bed temperatures, flow, speed, retraction & cooling are very important and need to be tuned for your specific printer and type of filament.





Recommended slicer settings

Below is a list of recommended slicer settings for a PLA+ Stormbird printed with a 0.4mm nozzle. These settings have been tested on a Creality Ender 3 S1 Pro & a Bambu Labs P1S with eSun PLA+ filament.

STORMBIRD V1 SLICER SETTINGS

Part Name	Layer Height	Line Width	Wall Line Count	Top Layers	Bottom Layers	Infill Density	Support Enabled	Brim Enabled
SINGLE WALL PARTS								
Canopy Front	0.2	0.42	1	3	3	4	Yes	No
Canopy Rear	0.2	0.42	1	3	3	4	No	No
Elevon Inner L	0.2	0.42	1	3	15	4	No	Yes
Elevon Inner R	0.2	0.42	1	3	15	4	No	Yes
Elevon Outer L	0.2	0.42	1	3	3	4	No	Yes
Elevon Outer R	0.2	0.42	1	3	3	4	No	Yes
Fuselage Mid	0.2	0.42	1	3	3	4	No	Yes
Fuselage Rear	0.2	0.42	1	3	3	4	No	Yes
Hatch	0.2	0.42	1	3	3	4	No	No
Nose	0.2	0.42	1	3	3	4	Yes	No
Servo Cover L	0.2	0.42	1	3	3	4	No	No
Servo Cover R	0.2	0.42	1	3	3	4	No	No
Tail Fin	0.2	0.42	1	3	3	4	No	No
Wing Inner L	0.2	0.42	1	3	3	4	Yes	Yes
Wing Inner R	0.2	0.42	1	3	3	4	Yes	Yes
Wing Outer L	0.2	0.42	1	3	3	4	No	Yes
Wing Outer R	0.2	0.42	1	3	3	4	No	Yes
Winglet L	0.2	0.42	1	3	3	4	No	Yes
Winglet R	0.2	0.42	1	3	3	4	No	Yes
SOLID PARTS								
Battery Tray	0.2	0.42	3	3	3	50	No	No
Flight Controller Tray	0.2	0.42	3	3	3	50	No	No
Motor Mount	0.2	0.42	3	3	3	50	No	No
Servo Tray (2)	0.2	0.42	3	3	3	50	No	No

Assembly

The assembly instructions are available in video format.

www.youtube.com/watch?v=6wLf6x558MM

1. Prepare the 3D printed parts

Print the Stormbird parts using the recommended slicer settings. Remove any leftover brim/support pieces and sand down rough edges.

2. Prepare the carbon fiber

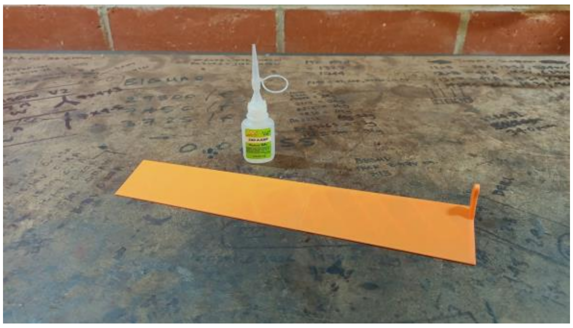
Cut the following carbon fiber tubes/rods:

- 6mm x 1m (1)
- 6mm x 300mm (1)
- 3mm x 330mm (2)
- 3mm x 30mm (3)

3. Join the elevon halves

Apply medium CA to the top of the elevon inner and join it together with the elevon outer.

Be careful not to get any glue inside the 3mm holes. A piece of 3mm carbon fiber tube can be inserted into the elevon to help with alignment as the glue dries but DO NOT glue it in place.



4. Join the wing halves

Apply medium CA to the top of the wing inner and join it together with the wing outer.

Be careful not to get any glue inside the 6mm spar hole. The 6mm x 1m carbon fiber tube can be inserted into the wing to help with alignment.

5. Attach the elevons

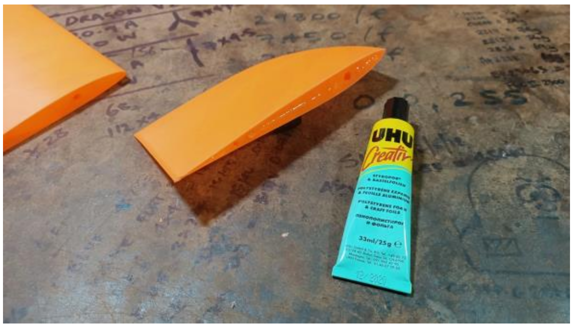
Align the elevon with the wing panel. Insert a 3mm x 330mm carbon fiber rod into the hole in the wing tip end. Push the rod all the way through the elevon and into the wing panel to lock the elevon in position.

Check the elevon moves freely. Sand the inside edges of the wing panel if needed.

6. Attach the winglets

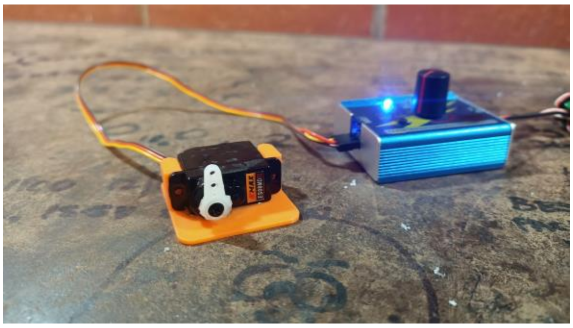
Apply UHU Por to the bottom of the winglet. Attach the winglet to the protruding 3mm carbon fiber rod and join it to the wing end.

The wings must be fully assembled with the winglet in place before attaching them to the fuselage.



7. Prepare the servos

Center the servo and attach a control horn at a 90-degree angle. Apply UHU Por to the bottom of the servo and attach it to the servo tray.



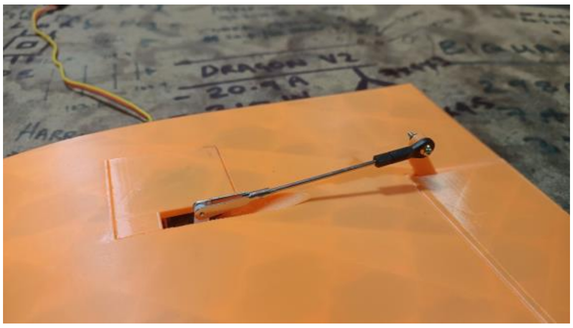
8. Install the servos

Apply UHU Por to the bottom of the servo tray and insert it into the wing. Feed the servo cables through the exit hole. Apply UHU Por to the top of the servo and attach the servo cover.

9. Install the linkages

Add a clevis and a ball link to a piece of threaded rod. Attach the clevis to the servo control arm and the ball link to the elevon control horn. Set the elevon with 1-2mm of up reflex.

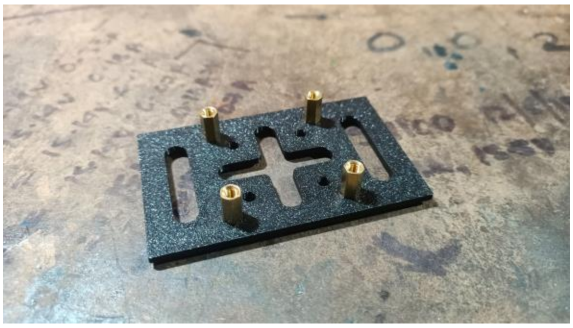
The linkage should be approximately 84mm from hole to hole.

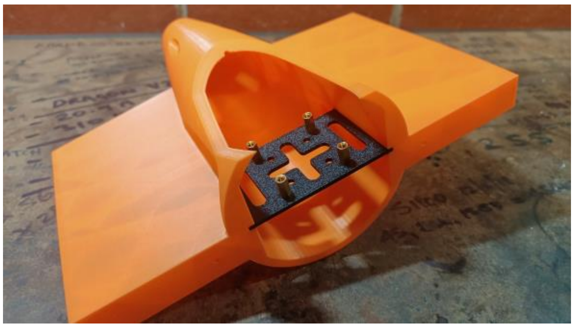


10. Install the flight controller tray

If needed, add spacers/standoffs to the flight controller tray. Insert the tray into the fuselage rear. Add a few drops of medium CA to hold it in place.

The flight controller tray must be inserted into the fuselage rear before joining the two main fuselage parts together.





11. Join the fuselage halves

Apply medium CA to the alignment holes in the fuselage mid. Insert pieces of filament into the holes and trim them with 5mm remaining. Apply CA to the end of the fuselage rear and join it together with the fuselage mid.

12. Attach the nose

Apply medium CA to the alignment holes in the nose. Insert pieces of filament into the holes and trim them with 5mm remaining. Apply UHU Por to the end of the nose and join it together with the fuselage mid.

The nose must be carefully aligned with the fuselage to keep the canopy rail system straight.

13. Install the battery plate

Apply UHU Por to the bottom of the battery plate and insert it into the fuselage.

14. Attach the tail fin

Apply medium CA into the holes underneath the tail fin. Insert two pieces of 3mm x 30mm carbon fiber rod into the holes. Apply UHU Por to the tail fin and attach it to the fuselage.

15. Install the fuselage magnets

Apply medium CA to the hole in the front of the nose and insert a 6mm x 3mm round magnet. Apply medium CA to the hole at the end of the fuselage mid and insert another magnet.

16. Attach the wing spars

Insert the 6mm x 1m carbon fiber tube into the front of the fuselage and the 6mm x 300mm tube into the rear.

The front spar extends 39cm from either side of the fuselage and the rear spar extends 4cm from either side.

17. Attach the wings

Connect the wings to the wing spars. Feed the servo cables through the fuselage holes. Apply UHU Por to the end of the wings and join them to the fuselage.

18. Prepare the motor & ESC

Connect/solder the motor wires to the ESC. Check for correct rotation direction.

19. Install the motor & ESC

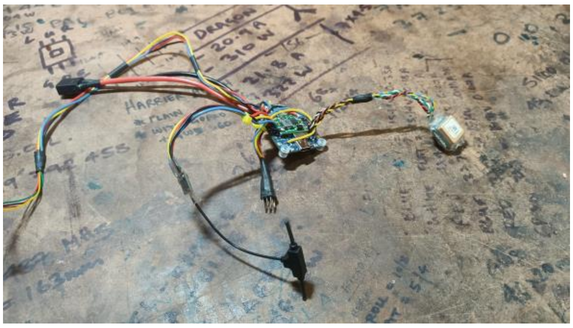
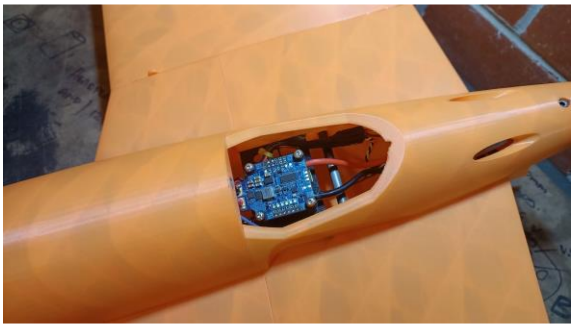
Attach the motor to the motor mount using four M3 x 8mm hex head screws. Press an M3 locknut into the bottom of the motor mount. Insert the ESC & motor mount into the rear of the fuselage. Attach the motor mount using an M3 x 10mm hex head screw.

The motor mount can be attached to the fuselage using glue only if the fuselage bolt is unusable.

20. Install the electronic gear

Install the flight controller, receiver, GPS, BEC, FPV gear, etc. Connect the servos and any other wiring.

21. Prepare the canopy

Apply medium CA to the alignment holes in the canopy front. Insert pieces of filament into the holes and trim them to a 5mm length. Apply CA to the end of the canopy front and join it together with the canopy rear. Check the polarity of the magnet in the fuselage nose. Apply medium CA to the magnet hole and insert a 6mm x 3mm round magnet.

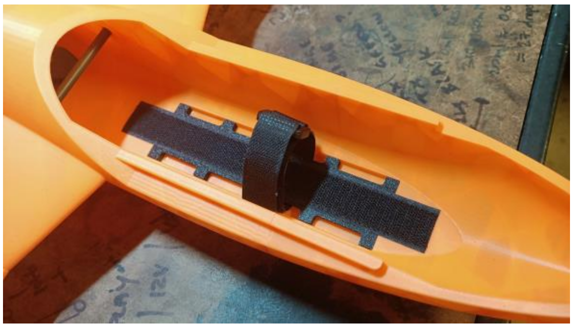
Allow the CA to fully dry before sliding the canopy on the fuselage rails.

22. Prepare the hatch

Apply medium CA to the hole at the end of the hatch. Insert a piece of 3mm x 30mm carbon fiber tube into the hole. Check the polarity of the magnet in the fuselage mid. Apply medium CA to the magnet hole and insert a 6mm x 3mm round magnet.

23. Install the battery strap & velcro

Add a battery strap and a 150mm length of Velcro to the battery plate.



24. Attach the canopy & hatch

Slide the canopy on to the fuselage rails. Insert the hatch into the rear of the fuselage.

Setup

Throw

Roll = 15mm up / 13mm down

Pitch = 6mm up / 5mm down

Reflex

1-2mm up

Center of gravity

The CG is 36mm behind the straight leading edge of the wing. There are CG bumps underneath the wing.

Notes

- Two different types of glue are used. Medium CA is used on the parts that need a strong permanent bond. A removeable glue such as UHU Por (or E6000) is used on parts that can be damaged then easily replaced, such as the winglets, nose, tail fin and the main join between the wings & fuselage.
- It is important to fully assemble the wings before attaching them to the fuselage. The winglets cannot be attached if the rest of the wings are already glued to the fuselage.
- Please contact me if you need any custom parts for FPV setups, I'm happy to help.